

30-Day ML + AWS Study Plan

Daily supervised + unsupervised practice alongside AWS Certified Machine Learning Engineer – Associate prep

Overview

Your 12 hours are split into three blocks: about 5 hours for the two ML problems (2.5 hours each), 6 to 7 hours for AWS, and the remainder for breaks and review.

A caution: running 12 hours a day for 30 days at full intensity tends to fall apart around week 3. Make one day a week (say Sunday) lighter, with only a review and a practice exam, and protect your sleep. A burned-out week 4 costs more than the lost hours.

Fixing the EDA and Cleaning Time Sink

- 1 **Put a 30-minute cap on EDA.** When the timer ends, build a baseline model anyway. The model will show you which features matter, and it is faster than staring at plots.
- 2 **Write your own EDA function once,** then reuse it every day. It should print shape, dtypes, missing %, cardinality, target distribution, skew of numeric columns, and top correlations with the target. Use a profiler like ydata-profiling for the first pass, and only chart what it flags.
- 3 **Ask questions instead of plotting everything.** Before each chart, write down what decision it will change (log-transform, drop, bin, encode). If it won't change a decision, skip it.
- 4 **Build a reusable preprocessing skeleton.** Keep a ColumnTransformer + Pipeline template with numeric, categorical, and high-cardinality branches, and only edit the column lists each day. Cleaning then becomes 15 minutes of configuration.
- 5 **Decide with cross-validation, not intuition.** Whether to impute, clip outliers, or log-transform is settled by comparing CV scores, not by debating it.
- 6 **Keep a "rules of thumb" notebook** with entries like "right-skewed target -> log" and "high-cardinality categorical -> target/frequency encoding". After about 10 problems, most of your EDA becomes recall.

Daily Template (2.5 hrs per problem)

Step	Time
EDA (hard cap)	25 min
Pipeline + baseline	25 min
Model improvement and tuning	55 min
Error analysis / cluster interpretation	25 min
Write-up: what worked, what didn't	20 min

30-Day Problem List

One supervised and one unsupervised problem per day (2.5 hrs each). Shaded rows are review days: redo the weakest problem of the week and update your rules-of-thumb notebook.

Day	Supervised	Unsupervised	AWS focus
Days 1-7: Classification Clustering			
1	Telco churn (logistic regression)	Mall customers (K-Means)	Course + hands-on labs
2	Heart disease (KNN, SVM)	Wholesale customers (scaling, silhouette)	Course + hands-on labs
3	Credit default (decision trees)	Wine (hierarchical clustering)	Course + hands-on labs
4	Bank marketing (imbalance)	Geospatial (DBSCAN)	Course + hands-on labs
5	Credit card fraud (extreme imbalance, PR-AUC)	Gaussian Mixture Models	Course + hands-on labs
6	Adult income (categoricals, fairness)	Online Retail (RFM segmentation)	Course + hands-on labs
7	REVIEW: redo weakest of days 1-6	REVIEW: redo weakest of days 1-6	Course + hands-on labs
Days 8-14: Regression Dimensionality reduction			
8	Ames house prices (regularization)	PCA on digits	Course + hands-on labs
9	Bike sharing (datetime features)	t-SNE and UMAP	Course + hands-on labs
10	Used car prices (skewed target)	PCA as preprocessing before regression	Course + hands-on labs
11	Insurance costs	Autoencoder on Fashion-MNIST	Course + hands-on labs
12	Energy efficiency	Unsupervised feature selection	Course + hands-on labs
13	Store sales (time-based split)	Image compression with SVD	Weak domains + practice exams
14	REVIEW: redo weakest of days 8-13	REVIEW: redo weakest of days 8-13	Weak domains + practice exams
Days 15-21: Boosting and harder cases Anomaly detection and recommenders			
15	Random Forest vs XGBoost vs LightGBM	Isolation Forest	Weak domains + practice exams
16	Hyperparameter tuning with Optuna	Local Outlier Factor (LOF)	Weak domains + practice exams
17	CatBoost on high-cardinality categoricals	One-Class SVM	Weak domains + practice exams
18	Forest cover type (multiclass)	Autoencoder anomaly detection	Weak domains + practice exams
19	SMS spam / news (TF-IDF)	Time-series anomalies	Weak domains + practice exams

Day	Supervised	Unsupervised	AWS focus
20	Multi-label classification	Market-basket analysis (Apriori)	Weak domains + practice exams
21	Probability calibration	MovieLens recommender (item-based, SVD)	Weak domains + practice exams
Days 22-30: Advanced and end-to-end Text, evaluation and capstone			
22	Stacking / blending	Topic modeling (LDA / NMF, 20 newsgroups)	Weak domains + practice exams
23	Time-series forecasting	Clustering text embeddings	Full timed mocks, patch gaps
24	Survival / churn-timing	Spectral clustering	Full timed mocks, patch gaps
25	Feature selection methods	Cluster stability and evaluation	Full timed mocks, patch gaps
26	Deploy a model to a SageMaker endpoint	Semi-supervised learning	Full timed mocks, patch gaps
27	End-to-end mock project 1	Capstone part 1: build	Full timed mocks, patch gaps
28	End-to-end mock project 2	Capstone part 2: refine and write up	Full timed mocks, patch gaps
29	REVIEW: redo weakest topic	REVIEW: redo weakest topic	Light review
30	Light review only	Light review only	Light review

AWS MLA-C01 Block (about 6–7 hrs/day)

- **Days 1–12:** Work through one structured course alongside hands-on labs in SageMaker (training jobs, Pipelines, Feature Store, Model Monitor), plus Glue, S3, Kinesis, and IAM basics.
- **Days 13–22:** Attack your weak domains, and take a practice exam every 3–4 days, reviewing every wrong answer.
- **Days 23–30:** Take full timed mocks, patch gaps, and do a light review the last two days.

As far as I know, the exam covers data preparation, model development, deployment and orchestration, and monitoring and security, with roughly equal weight. It is heavy on MLOps and AWS services, not algorithm theory, which is why your daily ML practice and the AWS study complement each other. Check the official exam guide to confirm the current domain weights.